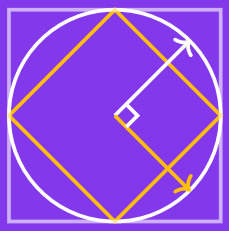
Linear Algebra, But Make It Make Sense

Lecture 5 – Lengths, angles & orthogonality (norms, projections, spectral theorem, SVD)



Sasha Sutton – personal study notes, Fall 2026

My notes + proofs + fully solved exercises, remixed into one doc.

How to read this doc (the box legend)

DEFINITION	The lore. Learn these word for word, examiners love them.
THEOREM	The canon. Things that are always true.
RECEIPTS	Proofs. Every claim needs receipts.
EXAMPLE	Worked examples with actual numbers.
ML SIDE QUEST	Why a data scientist should care.
EXAM CHEAT CODE	"If they ask you *this*, you do *that*". Step-by-step recipes.
COMMON L	Classic mistakes. Don't be that person.
TL;DR	The summary for when you're cramming at 2am.

Everything here is real maths, just explained like a friend would. The vibes are optional, the rigour is not.

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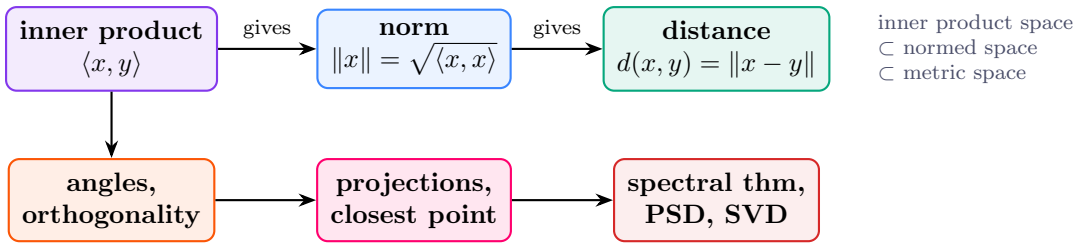
1 The big picture (read this first)

Previously on Linear Algebra

- **Lectures 1–2:** vector spaces, linear maps, matrices, rank, kernel, linear systems.
- **Lecture 3:** determinants. **Lecture 4:** eigenvalues, diagonalisation.
- **Today:** so far our vectors had *no length* and *no angles*. We add a tool that measures both. Everything else today grows out of that one tool.

TL;DR // The whole lecture as a story

1. An **inner product** $\langle x, y\rangle$ is a "multiply two vectors, get a number" machine (the dot product is the classic one).
2. From it you get a **length** (the **norm** $\|x\| = \sqrt{\langle x, x\rangle}$), then a **distance** ($\|x - y\|$), and an **angle** (via $\cos\theta$).
3. Two vectors are **orthogonal** (perpendicular) when $\langle x, y\rangle = 0$. That gives Pythagoras, Cauchy–Schwarz, and cosine similarity.
4. **Orthogonal projection** = the closest point of a subspace. It's how you "drop a perpendicular" in any dimension (least squares, PCA).
5. **Spectral theorem:** a *symmetric* matrix can always be diagonalised with *perpendicular* eigenvectors: $A = VDV^\top$.
6. **Positive (semi-)definite** matrices: symmetric matrices with eigenvalues > 0 (≥ 0). Covariance and Gram matrices are like that.
7. **SVD:** *every* matrix (even rectangular) is rotate → stretch → rotate: $A = U\Sigma V^\top$.



Not every distance comes from a norm, and not every norm comes from an inner product. But each level gives you the next one for free.

2 Inner products: multiplying two vectors

2.1 The dot product you already know

DEFINITION // Dot product (scalar product) on $\mathbb{R}^n$

For $x = (x_1, \ldots, x_n)$ and $y = (y_1, \ldots, y_n)$ in $\mathbb{R}^n$:

$$\langle x, y\rangle = x \cdot y = \sum_{i=1}^{n} x_i y_i = x^\top y.$$

The last form is a matrix product: a $1 \times n$ row times an $n \times 1$ column gives a 1×1 matrix, i.e.

a number.

EXAMPLE // Computing dot products

1. $x = (1, 2, 3)$, $y = (4, -1, 2)$: $\langle x, y\rangle = 1 \cdot 4 + 2 \cdot (-1) + 3 \cdot 2 = 4 - 2 + 6 = 8$.
2. $x = (1, 1)$, $y = (1, -1)$: $\langle x, y\rangle = 1 - 1 = 0$. These two are **perpendicular** (draw them!).
3. $\langle x, x\rangle = x_1^2 + \cdots + x_n^2$: always ≥ 0. For $x = (3, 4)$: $9 + 16 = 25 = 5^2$, the squared length.

2.2 The general definition

The dot product is not the only useful "product". Here are the three rules that make something an inner product.

DEFINITION // Inner product

An **inner product** on a real vector space V is a map $\langle \cdot, \cdot \rangle : V \times V \to \mathbb{R}$ such that for all $x, y, z \in V$, $\alpha, \beta \in \mathbb{R}$:

(i) **Symmetric:** $\langle x, y\rangle = \langle y, x\rangle$.

(ii) **Bilinear:** $\langle \alpha x + \beta y, z\rangle = \alpha\langle x, z\rangle + \beta\langle y, z\rangle$ (linear in the first slot; with symmetry, also in the second).

(iii) **Positive-definite:** $\langle x, x\rangle \geq 0$, and $\langle x, x\rangle = 0 \iff x = 0_V$.

A vector space with an inner product is an **inner product space** (FR: *espace préhilbertien*). If it is also finite-dimensional, it is a **Euclidean space**.

EXAM CHEAT CODE // "Show that φ is an inner product"

Check the three rules, **in this order**:

1. **Symmetry:** swap the two arguments in the formula; show you get the same thing.
2. **Linearity in the first argument:** plug $\alpha x + \beta y$ in the first slot and expand. (Then say: "by symmetry, also linear in the second".)
3. **Positivity:** show $\varphi(x, x) \geq 0$ (usually a sum of squares, or an integral of something ≥ 0).
4. **Definiteness** (the hard part): assume $\varphi(x, x) = 0$ and prove $x = 0$.

EXAMPLE // Three inner products you should know

1. Weighted dot product on $\mathbb{R}^n$. For fixed weights $\alpha_1, \ldots, \alpha_n > 0$: $\langle x, y\rangle_\alpha = \sum_{i=1}^n \alpha_i x_i y_i$. Definiteness: $\sum \alpha_i x_i^2 = 0$ with all $\alpha_i > 0$ forces every $x_i = 0$. (If some α_i were 0 or negative, it would fail: e.g. $\alpha = (1, 0)$ gives $\langle (0, 1), (0, 1)\rangle_\alpha = 0$ with $(0, 1) \neq 0$.)

2. Functions. On $E = C^0([a, b], \mathbb{R})$ (continuous functions): $(f \mid g) = \int_a^b f(x)g(x)\,dx$.

- Symmetric and bilinear: because the integral is linear and $fg = gf$.
- Positive: $(f \mid f) = \int_a^b f^2 \geq 0$.
- Definite: f^2 is continuous and ≥ 0; a continuous non-negative function with integral 0 is identically 0. So $f = 0$.

3. Matrices. On $\mathcal{M}_n(\mathbb{R})$: $(A \mid B) = \operatorname{tr}(A^\top B)$. In fact $\operatorname{tr}(A^\top B) = \sum_{i,j} a_{ij} b_{ij}$: it's just the dot product of the two matrices seen as long vectors of n^2 numbers. So $\operatorname{tr}(A^\top A) = \sum_{i,j} a_{ij}^2 \geq 0$, and $= 0$ only if all $a_{ij} = 0$. (Full proof in the exercise solutions.)

THEOREM / PROPOSITION // Moving a matrix across the dot product

For $A \in \mathbb{R}^{m\times n}$, $x \in \mathbb{R}^n$, $y \in \mathbb{R}^m$: $\langle Ax, y\rangle = \langle x, A^\top y\rangle$.

RECEIPTS (proof) //

$\langle Ax, y\rangle = (Ax)^\top y = x^\top A^\top y = \langle x, A^\top y\rangle$, using $(Ax)^\top = x^\top A^\top$ (Lecture 2). ■ *(and that's on periodt)*

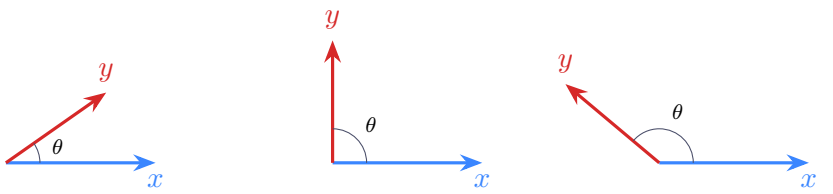
2.3 Orthogonal vectors and angles

DEFINITION // Orthogonal

x and y are **orthogonal**, written $x \perp y$, if $\langle x, y\rangle = 0$. This generalises "perpendicular" from 2D to any dimension (and even to functions or matrices).

THEOREM / PROPOSITION // The angle formula

For non-zero $x, y \in \mathbb{R}^n$, with θ the angle between them: $\langle x, y\rangle = \|x\|_2 \, \|y\|_2 \cos\theta$.



acute: $\langle x, y\rangle > 0$ right angle: $\langle x, y\rangle = 0$ obtuse: $\langle x, y\rangle < 0$

THEOREM / PROPOSITION // Orthogonal non-zero vectors are independent

If $v_1, \ldots, v_p$ are **non-zero** and pairwise orthogonal, they are linearly independent (so $p \le n$ in $\mathbb{R}^n$).

RECEIPTS (proof) // the dot-product trick

Suppose $\alpha_1 v_1 + \cdots + \alpha_p v_p = 0$. Fix j and take the inner product of both sides with v_j:

$$\sum_{i=1}^{p} \alpha_i \langle v_i, v_j\rangle = \langle 0, v_j\rangle = 0.$$

Every term with $i \neq j$ is 0 (orthogonal), so only $\alpha_j \langle v_j, v_j\rangle = \alpha_j \|v_j\|^2$ survives. Since $v_j \neq 0$, $\|v_j\|^2 > 0$, hence $\alpha_j = 0$. This works for every j. ■ *(and that's on periodt)*

This "dot both sides with v_j" move is the single most useful trick of the lecture. You'll use it again and again.

3 Distances (metric spaces)

DEFINITION // Metric (distance)

A **metric** on a set S is a function $d : S \times S \to \mathbb{R}_+$ such that for all $x, y, z \in S$:

(i) **Positivity:** $d(x, y) \geq 0$, and $d(x, y) = 0 \iff x = y$;

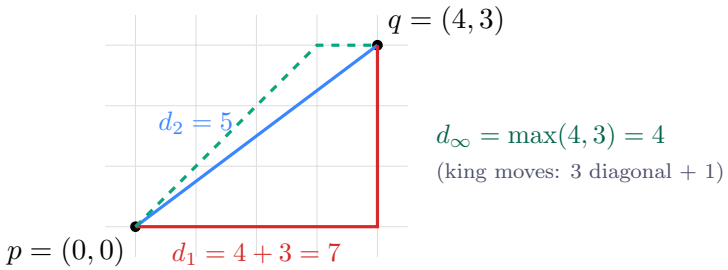
(ii) **Symmetry:** $d(x, y) = d(y, x)$;

(iii) **Triangle inequality:** $d(x, z) \leq d(x, y) + d(y, z)$ *(a detour is never shorter)*.

(S, d) is a **metric space**. S doesn't need to be a vector space: it can be words, graphs, anything.

EXAMPLE // Metrics in the wild

1. On $\mathbb{R}$: $d(x, y) = |y - x|$.
2. **Hamming distance** on words of length n: number of positions where they differ. $d(\texttt{karolin}, \texttt{kathrin}) = 3$ (positions 3, 4, 5). Used in information theory to count transmission errors.
3. On $\mathbb{R}^2$, between $p = (x_1, y_1)$ and $q = (x_2, y_2)$:
   - **Euclidean:** $d_2(p, q) = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$ (as the crow flies);
   - **Manhattan / taxi-cab:** $d_1(p, q) = |x_2 - x_1| + |y_2 - y_1|$ (walking along a street grid);
   - **Max:** $d_\infty(p, q) = \max\{|x_2 - x_1|, |y_2 - y_1|\}$ (a chess king's number of moves);
   - **Discrete:** $d(p, q) = 0$ if $p = q$, 1 otherwise.



ML SIDE QUEST // The metric decides who your nearest neighbour is

A nearest-neighbour classifier predicts with the closest training point: $i^\star = \arg\min_i d(x_q, x^{(i)})$. **Example:** query $x_q = (0, 0)$, two training points $a = (3, 0)$ and $b = (2, 2)$.

metric	$d(x_q, a)$	$d(x_q, b)$	nearest
d_2	3	$\sqrt{8} \approx 2.83$	b
d_1	3	4	a
d_∞	3	2	b

Same data, different metric, different prediction. Euclidean measures geometric closeness; Manhattan often suits sparse or grid-like features; Hamming compares binary codes. **Changing the metric changes what the algorithm calls "similar".**

TL;DR // Why metrics matter beyond ML

A metric lets you talk about **limits** for things that aren't numbers: $x_n \to x$ means $d(x_n, x) \to 0$, i.e. for every $\varepsilon > 0$ there is N with $d(x_n, x) < \varepsilon$ for all $n \geq N$.

4 Norms: the length of a vector

4.1 Definition

DEFINITION // Norm

A **norm** on a real vector space V is a function $\|\cdot\| : V \to \mathbb{R}_+$ such that for all $x, y \in V$, $\alpha \in \mathbb{R}$:

(i) **Positivity:** $\|x\| \geq 0$, and $\|x\| = 0 \iff x = 0_V$;

(ii) **Homogeneity:** $\|\alpha x\| = |\alpha| \, \|x\|$ (twice as long a vector, twice the length; and $\|-x\| = \|x\|$);

(iii) **Triangle inequality:** $\|x + y\| \leq \|x\| + \|y\|$.

$(V, \|\cdot\|)$ is a **normed space**. Every norm gives a distance: $d(x, y) = \|x - y\|$.

THEOREM / PROPOSITION // An inner product gives a norm

If $\langle \cdot, \cdot \rangle$ is an inner product, then $\|x\| = \sqrt{\langle x, x \rangle}$ is a norm (the **induced norm**). So every inner product space is a normed space, hence a metric space.

Positivity and homogeneity are immediate ($\langle \alpha x, \alpha x \rangle = \alpha^2 \langle x, x \rangle$). The triangle inequality needs Cauchy–Schwarz: proved below.

4.2 The norms of $\mathbb{R}^n$

DEFINITION // L^1, L^2, L^p, L^∞ norms

For $x = (x_1, \ldots, x_n) \in \mathbb{R}^n$:

$$\|x\|_1 = \sum_{i=1}^{n} |x_i|, \qquad \|x\|_2 = \sqrt{\sum_{i=1}^{n} x_i^2}, \qquad \|x\|_p = \Big(\sum_{i=1}^{n} |x_i|^p \Big)^{1/p} \ (p \geq 1), \qquad \|x\|_\infty = \max_{1 \leq i \leq n} |x_i|.$$

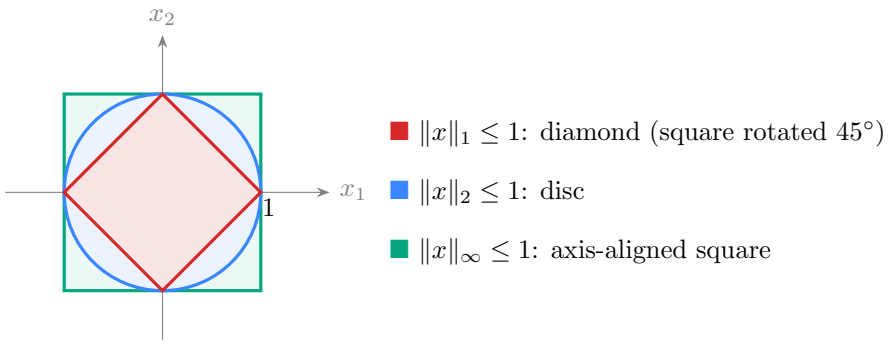
$\|\cdot\|_2$ is the **Euclidean norm**, induced by the dot product: $\langle x, x \rangle = \|x\|_2^2$. The 1- and 2-norms are p-norms; $\|\cdot\|_\infty$ is the limit of $\|\cdot\|_p$ when $p \to \infty$. We need $p \geq 1$: for $0 < p < 1$ the triangle inequality fails.

EXAMPLE // One vector, three lengths

$x = (3, -4)$:

1. $\|x\|_1 = |3| + |-4| = 7$.
2. $\|x\|_2 = \sqrt{9 + 16} = 5$.
3. $\|x\|_\infty = \max(3, 4) = 4$.

Always $\|x\|_\infty \leq \|x\|_2 \leq \|x\|_1$. **Why $p \geq 1$:** for "$p = \frac{1}{2}$", $x = (1, 0)$ and $y = (0, 1)$ have "length" 1 each, but $x + y = (1, 1)$ gets $(1 + 1)^2 = 4 > 1 + 1$: the triangle inequality breaks.



The **unit ball** $\{x : \|x\| \leq 1\}$ for three norms of $\mathbb{R}^2$. Bigger p = fatter ball.

4.3 Pythagoras and Cauchy–Schwarz

THEOREM / PROPOSITION // Expanding a squared norm

In any inner product space: $\|x+y\|^2 = \|x\|^2 + 2\langle x, y\rangle + \|y\|^2$.

RECEIPTS (proof) //

$\|x+y\|^2 = \langle x+y, x+y\rangle = \langle x,x\rangle + \langle x,y\rangle + \langle y,x\rangle + \langle y,y\rangle = \|x\|^2 + 2\langle x,y\rangle + \|y\|^2$ (bilinearity, then symmetry). ■ *(and that's on periodt)*

THEOREM / PROPOSITION // Pythagorean theorem

If $x \perp y$, then $\|x+y\|^2 = \|x\|^2 + \|y\|^2$.

Proof: the previous formula with $\langle x, y\rangle = 0$. Works in any inner product space: for vectors of $\mathbb{R}^{1000}$, for functions, for matrices.

THEOREM / PROPOSITION // Cauchy–Schwarz inequality

In any inner product space, for all x, y: $|\langle x, y\rangle| \leq \|x\|\,\|y\|$.
Equality holds **iff** x and y are linearly dependent (one is a multiple of the other, including when one is 0).

RECEIPTS (proof) // remove the part of x along y

If $y = 0$ both sides are 0. Otherwise let $z = x - \dfrac{\langle x, y\rangle}{\|y\|^2} y$ ("x minus its shadow on y"). Then:

1. $0 \leq \|z\|^2$.
2. Expand: $\|z\|^2 = \|x\|^2 - 2\dfrac{\langle x,y\rangle}{\|y\|^2}\langle x,y\rangle + \dfrac{\langle x,y\rangle^2}{\|y\|^4}\|y\|^2 = \|x\|^2 - \dfrac{\langle x,y\rangle^2}{\|y\|^2}$.
3. So $\langle x,y\rangle^2 \leq \|x\|^2\|y\|^2$; take square roots.
4. Equality $\iff$ $z = 0$ $\iff$ x is a multiple of y. ■ *(and that's on periodt)*

RECEIPTS (proof) // the triangle inequality for an induced norm

$\|x+y\|^2 = \|x\|^2 + 2\langle x,y\rangle + \|y\|^2 \leq \|x\|^2 + 2\|x\|\|y\| + \|y\|^2 = (\|x\| + \|y\|)^2$, using Cauchy–Schwarz on the middle term. Take square roots. ■ *(and that's on periodt)*

ML SIDE QUEST // Cosine similarity

$$\text{cos-sim}(x,y) = \frac{\langle x,y\rangle}{\|x\|_2\|y\|_2} = \cos\theta \ \in [-1,1] \quad \text{(by Cauchy–Schwarz)}.$$

1 = same direction, 0 = orthogonal (unrelated), -1 = opposite. It ignores length and only compares *direction*: the standard way to compare embeddings in retrieval and self-supervised learning.
Example: $x = (1,2,0)$, $y = (2,1,2)$: $\langle x,y\rangle = 2+2+0 = 4$, $\|x\| = \sqrt{5}$, $\|y\| = 3$, so cos-sim $= \frac{4}{3\sqrt{5}} \approx 0.60$ (angle $\approx 53°$).

ML SIDE QUEST // Norms as regularisers

Training often minimises $\text{loss}(w) + \lambda\|w\|$: "fit the data, but keep w small". Which norm you pick decides what "small" means:

- $\|w\|_2^2$ (ridge / weight decay): penalises big weights smoothly, shrinks everything a bit;
- $\|w\|_1$ (lasso): pushes many weights to **exactly** 0: sparse models (the diamond's corners sit on the axes);
- $\|w\|_\infty$: controls the largest coordinate.

4.4 Bonus: equivalence of norms, matrix norms

THEOREM / PROPOSITION // All norms are equivalent in finite dimension (bonus)

If $\dim V < \infty$, for any two norms $\|\cdot\|_A, \|\cdot\|_B$ there are constants $\alpha, \beta > 0$ with $\alpha\|x\|_A \le \|x\|_B \le \beta\|x\|_A$ for all x. So "$x_n \to x$" means the same thing for every norm. (False in infinite dimension, e.g. for function spaces.) Example in $\mathbb{R}^n$: $\|x\|_\infty \le \|x\|_2 \le \sqrt{n}\,\|x\|_\infty$.

DEFINITION // Matrix norms (bonus)

- **Entry-wise** norms: treat $A \in \mathbb{R}^{m\times n}$ as a vector of mn numbers. For $p = 2$: the **Frobenius norm** $\|A\|_F = \sqrt{\sum_{i,j} a_{ij}^2} = \sqrt{\text{tr}(A^\top A)}$.
- **Induced** (operator) norms: how much A can stretch a vector:

$$\|A\| = \sup_{x\neq 0} \frac{\|Ax\|}{\|x\|} = \sup_{\|x\|\le 1} \|Ax\|.$$

  With p-norms: $\|A\|_1$ = max absolute **column** sum; $\|A\|_\infty$ = max absolute **row** sum; $\|A\|_2 = \sqrt{\lambda_{\max}(A^\top A)} = \sigma_{\max}(A)$, the **spectral norm** (Section 8). Always $\|A\|_2 \le \|A\|_F$.

EXAMPLE // All four on one matrix

$A = \begin{pmatrix} 1 & -2 \\ 3 & 4 \end{pmatrix}$.

1. $\|A\|_1$: column sums $|1|+|3| = 4$ and $|-2|+|4| = 6$, so 6.
2. $\|A\|_\infty$: row sums $1+2 = 3$ and $3+4 = 7$, so 7.
3. $\|A\|_F = \sqrt{1+4+9+16} = \sqrt{30} \approx 5.48$.

4. $\|A\|_2$: $A^\top A = \begin{pmatrix} 10 & 10 \\ 10 & 20 \end{pmatrix}$, $\chi = \lambda^2 - 30\lambda + 100$, $\lambda_{\max} = 15 + \sqrt{125} \approx 26.18$, so $\|A\|_2 \approx 5.12 \leq 5.48$ ✓.

ML SIDE QUEST // Frobenius norm = total squared error

For a data matrix X and a reconstruction $\widehat{X}$ (PCA, matrix factorisation, autoencoders, denoising): $\|X - \widehat{X}\|_F^2 = \sum_{i,j}(X_{ij} - \widehat{X}_{ij})^2$, the total squared reconstruction error over all entries.

5 Orthonormal bases, orthogonal complements, projections

TL;DR // Chapter 5 in one breath

An **orthonormal basis** is the dream basis: perpendicular unit vectors, so coordinates are just dot products. The **orthogonal complement** $F^\perp$ is everything perpendicular to F. The **orthogonal projection** of x onto F is the point of F closest to x: you get it by "dropping a perpendicular".

5.1 Orthonormal bases

DEFINITION // Orthogonal / orthonormal basis

A basis $(e_1, \ldots, e_n)$ of an inner product space is **orthogonal** if $\langle e_i, e_j \rangle = 0$ for $i \neq j$, and **orthonormal** if moreover every $\|e_i\| = 1$. Compactly: $\langle e_i, e_j \rangle = \delta_{ij}$ (1 if $i = j$, 0 otherwise).

THEOREM / PROPOSITION // Coordinates are dot products

If $(e_1, \ldots, e_n)$ is orthonormal, every x satisfies $x = \sum_{i=1}^n \langle x, e_i \rangle e_i$, and $\|x\|^2 = \sum_{i=1}^n \langle x, e_i \rangle^2$.

RECEIPTS (proof) // dot both sides with e_j

Write $x = \sum_i c_i e_i$. Then $\langle x, e_j \rangle = \sum_i c_i \langle e_i, e_j \rangle = c_j$. For the norm: $\|x\|^2 = \langle \sum_i c_i e_i, \sum_j c_j e_j \rangle = \sum_{i,j} c_i c_j \delta_{ij} = \sum_i c_i^2$. ■ *(and that's on periodt)*

Compare with Lecture 1: in a general basis, finding coordinates means solving a linear system. In an orthonormal basis it's one dot product per coordinate.

THEOREM / PROPOSITION // Existence

Every finite-dimensional inner product space has an orthonormal basis. (Constructed by Gram–Schmidt below.)

EXAM CHEAT CODE // "Find an orthonormal basis of $F = \operatorname{span}(w_1, \ldots, w_k)$" (Gram–Schmidt)

Turn independent vectors into orthonormal ones, one at a time:

1. $e_1 = \dfrac{w_1}{\|w_1\|}$.
2. For the next vector, **remove its components along the e's already built**, then nor-

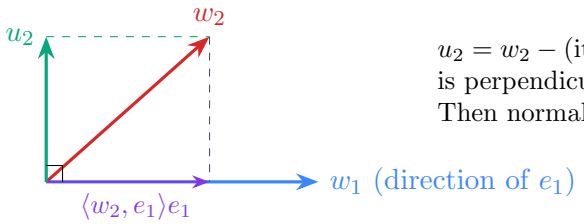
malise:

$$u_k = w_k - \sum_{i<k} \langle w_k, e_i \rangle \, e_i, \qquad e_k = \frac{u_k}{\|u_k\|}.$$

3. Check: all $\langle e_i, e_j \rangle = 0$ and all $\|e_i\| = 1$.

EXAMPLE // Gram–Schmidt on $w_1 = (1,1,0)$, $w_2 = (1,0,1)$

1. $\|w_1\| = \sqrt{2}$, so $e_1 = \frac{1}{\sqrt{2}}(1,1,0)$.
2. $\langle w_2, e_1 \rangle = \frac{1}{\sqrt{2}}(1+0+0) = \frac{1}{\sqrt{2}}$. Remove it: $u_2 = (1,0,1) - \frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}}(1,1,0) = (1,0,1) - (\frac{1}{2}, \frac{1}{2}, 0) = (\frac{1}{2}, -\frac{1}{2}, 1)$.
3. $\|u_2\| = \sqrt{\frac{1}{4} + \frac{1}{4} + 1} = \sqrt{\frac{3}{2}}$, so $e_2 = \frac{1}{\sqrt{6}}(1,-1,2)$.
4. Check: $\langle e_1, e_2 \rangle = \frac{1}{\sqrt{12}}(1-1+0) = 0$ ✓, $\|e_2\|^2 = \frac{1+1+4}{6} = 1$ ✓.



$u_2 = w_2 -$ (its shadow on e_1) is perpendicular to e_1. Then normalise: $e_2 = u_2/\|u_2\|$.

5.2 Orthogonal complement

DEFINITION // Orthogonal complement

For a subspace F of an inner product space E:

$$F^\perp = \{x \in E \mid \langle x, v \rangle = 0 \text{ for all } v \in F\}.$$

If $F = \operatorname{span}(f_1, \ldots, f_k)$, it's enough to check the spanning vectors: $x \in F^\perp \iff \langle x, f_1 \rangle = \cdots = \langle x, f_k \rangle = 0$.

EXAM CHEAT CODE // "Compute $F^\perp$"

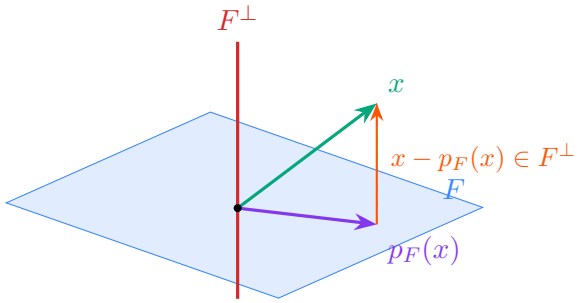
1. Take a spanning family $f_1, \ldots, f_k$ of F.
2. Write one equation per vector: $\langle x, f_i \rangle = 0$. That's a **homogeneous linear system** whose rows are the f_i.
3. Solve it (Lecture 2). The solution set is $F^\perp$; give a basis.
4. Check: $\dim F + \dim F^\perp = \dim E$, and each basis vector is orthogonal to each f_i.

In matrix terms: if M has rows $f_1, \ldots, f_k$, then $F^\perp = \ker M$.

EXAMPLE // Orthogonal complements in $\mathbb{R}^3$

(a) A line. $F = \operatorname{span}\big((1,1,0)\big)$. Equation: $x + y = 0$ (z free). $F^\perp = \operatorname{span}\big((1,-1,0),(0,0,1)\big)$: a plane. $1 + 2 = 3$ ✓.
(b) A plane. $F = \operatorname{span}\big((1,1,0),(1,0,1)\big)$. Equations $x + y = 0$ and $x + z = 0$, so $y = -x$,

$z = -x$: $F^\perp = \operatorname{span}\big((1,-1,-1)\big)$, the *normal* line. $2 + 1 = 3$ ✓.



A plane F, its orthogonal complement (the normal line), and the orthogonal projection of x onto F.

THEOREM / PROPOSITION // Properties of $F^\perp$

For a subspace F of a finite-dimensional inner product space E:

(i) $F^\perp$ is a subspace of E;

(ii) $F \cap F^\perp = \{0_E\}$;

(iii) $\dim F + \dim F^\perp = \dim E$ (so $E = F \oplus F^\perp$);

(iv) $(F^\perp)^\perp = F$.

(Proofs: the exercises, Question 5, in Section 9.)

5.3 Projections

DEFINITION // Projector, orthogonal projection

A linear map $p \in \mathcal{L}(E)$ is a **projector** if $p \circ p = p$ (projecting twice = projecting once).
The **orthogonal projection** onto a subspace F is the linear map $p_F : E \to E$ such that, for all x:

$$p_F(x) \in F \qquad \text{and} \qquad x - p_F(x) \in F^\perp.$$

THEOREM / PROPOSITION // Projection formula

If $(e_1, \dots, e_k)$ is an **orthonormal** basis of F, then

$$p_F(x) = \sum_{i=1}^{k} \langle x, e_i \rangle \, e_i.$$

RECEIPTS (proof) // check the two conditions

$p_F(x)$ is a combination of the e_i, so it's in F. For each j: $\langle x - p_F(x), e_j \rangle = \langle x, e_j \rangle - \sum_i \langle x, e_i \rangle \langle e_i, e_j \rangle = \langle x, e_j \rangle - \langle x, e_j \rangle = 0$, so $x - p_F(x)$ is orthogonal to a basis of F, hence to all of F. ■ *(and that's on periodt)*

COMMON L (don't do this) // Orthonormal, or it doesn't work

The formula $\sum \langle x, e_i \rangle e_i$ needs an **orthonormal** basis. With a merely orthogonal basis use $\sum \frac{\langle x, e_i \rangle}{\|e_i\|^2} e_i$; with a non-orthogonal one, run Gram–Schmidt first (or solve the normal equations).

EXAM CHEAT CODE // "Project x onto F" and "distance from x to F"

1. Get an orthonormal basis $(e_1, \dots, e_k)$ of F (Gram–Schmidt if needed).
2. Compute the coefficients $\langle x, e_i \rangle$.
3. $p_F(x) = \sum_i \langle x, e_i \rangle e_i$.
4. Residual $r = x - p_F(x)$: check $r \perp$ each e_i.
5. $d(x, F) = \|r\|$.

Shortcut when $F^\perp$ is a line $\operatorname{span}(n)$ (a hyperplane, e.g. a plane in $\mathbb{R}^3$): $d(x, F) = \dfrac{|\langle x, n \rangle|}{\|n\|}$ and $p_F(x) = x - \dfrac{\langle x, n \rangle}{\|n\|^2} n$.

EXAMPLE // Projecting $x = (1, 2, 3)$ onto $F = \operatorname{span}\big((1, 1, 0), (1, 0, 1)\big)$

Orthonormal basis from Gram–Schmidt above: $e_1 = \frac{1}{\sqrt{2}}(1, 1, 0)$, $e_2 = \frac{1}{\sqrt{6}}(1, -1, 2)$.

1. $\langle x, e_1 \rangle = \frac{1+2}{\sqrt{2}} = \frac{3}{\sqrt{2}}$, so the first piece is $\frac{3}{2}(1, 1, 0)$.
2. $\langle x, e_2 \rangle = \frac{1-2+6}{\sqrt{6}} = \frac{5}{\sqrt{6}}$, so the second piece is $\frac{5}{6}(1, -1, 2)$.
3. $p_F(x) = (\frac{3}{2} + \frac{5}{6},\ \frac{3}{2} - \frac{5}{6},\ \frac{5}{3}) = (\frac{7}{3}, \frac{2}{3}, \frac{5}{3})$.
4. $r = x - p_F(x) = (-\frac{4}{3}, \frac{4}{3}, \frac{4}{3}) = -\frac{4}{3}(1, -1, -1)$: a multiple of the normal $n = (1, -1, -1)$, so it's in $F^\perp$ ✓.
5. $d(x, F) = \|r\| = \frac{4}{3}\sqrt{3} \approx 2.31$. Shortcut check: $\frac{|\langle x, n \rangle|}{\|n\|} = \frac{|1-2-3|}{\sqrt{3}} = \frac{4}{\sqrt{3}}$ ✓.

THEOREM / PROPOSITION // Projections shrink (Bessel's inequality)

For all x: $\|p_F(x)\| \le \|x\|$, with equality iff $x \in F$.

RECEIPTS (proof) // Pythagoras

$x = p_F(x) + (x - p_F(x))$ with the two pieces orthogonal, so $\|x\|^2 = \|p_F(x)\|^2 + \|x - p_F(x)\|^2 \ge \|p_F(x)\|^2$. Equality iff $x - p_F(x) = 0$, i.e. $x = p_F(x) \in F$. ■ *(and that's on periodt)*

DEFINITION // Distance to a subspace

$d(x, F) = \inf_{y \in F} \|x - y\|$: the shortest distance from x to a point of F.

THEOREM / PROPOSITION // The projection is the closest point

$p_F(x)$ is the **unique** point of F with $\|x - p_F(x)\| = d(x, F)$.

RECEIPTS (proof) // Pythagoras again

For any $y \in F$: $x - y = \underbrace{(x - p_F(x))}_{\in F^\perp} + \underbrace{(p_F(x) - y)}_{\in F}$. These are orthogonal, so $\|x - y\|^2 = \|x - p_F(x)\|^2 + \|p_F(x) - y\|^2 \ge \|x - p_F(x)\|^2$, with equality iff $y = p_F(x)$. ■ *(and that's on periodt)*

ML SIDE QUEST // Least squares and PCA are projections

- **Linear regression:** the best fit $X\hat{\beta}$ is the orthogonal projection of y onto the column space of X. "Residual $\perp$ columns" is exactly $X^\top(y - X\hat{\beta}) = 0$, the normal equations of Lecture 2.
- **PCA** (Lecture 1): $\hat{x} = \sum_j (u_j^\top x)u_j$ is the projection formula with the orthonormal principal directions u_j. Reconstruction error = distance to the subspace.

THEOREM / PROPOSITION // Characterising orthogonal projectors

For a projector p ($p \circ p = p$) with $F = \text{range}(p)$, the following are equivalent:

(i) p is the orthogonal projection onto F (and then $\ker p = F^\perp$);

(ii) for all x, y: $\langle p(x), y - p(y)\rangle = 0$;

(iii) for all x, y: $\langle p(x), y\rangle = \langle x, p(y)\rangle$ (p is *symmetric*).

(Proof: the exercises, Question 7.)

5.4 Orthogonal maps and matrices

DEFINITION // Orthogonal endomorphism

$f \in \mathcal{L}(E)$ is **orthogonal** if it preserves the inner product: $\langle f(x), f(y)\rangle = \langle x, y\rangle$ for all x, y. Think: rotations and reflections (they keep lengths and angles).

THEOREM / PROPOSITION // Preserving lengths is enough

f is orthogonal $\iff$ $\|f(x)\| = \|x\|$ for all x.

RECEIPTS (proof) // polarisation identity

($\Rightarrow$) $\|f(x)\|^2 = \langle f(x), f(x)\rangle = \langle x, x\rangle = \|x\|^2$. ($\Leftarrow$) From the expansion formula, $\|x + y\|^2 - \|x - y\|^2 = 4\langle x, y\rangle$ (**polarisation**): inner products can be recovered from norms. So $4\langle x, y\rangle = \|x+y\|^2 - \|x-y\|^2 = \|f(x+y)\|^2 - \|f(x-y)\|^2 = \|f(x)+f(y)\|^2 - \|f(x)-f(y)\|^2 = 4\langle f(x), f(y)\rangle$ (linearity of f in the third step). ■ *(and that's on periodt)*

DEFINITION // Orthogonal matrix (again)

$Q \in \mathcal{M}_n(\mathbb{R})$ is **orthogonal** if $x \mapsto Qx$ is an orthogonal endomorphism. Equivalently: $Q^\top Q = QQ^\top = I_n$, i.e. $Q^{-1} = Q^\top$, i.e. the columns of Q form an orthonormal basis of $\mathbb{R}^n$. It preserves dot products: $(Qx)^\top(Qy) = x^\top Q^\top Qy = x^\top y$. And $\det Q = \pm 1$ (Lecture 3).

EXAMPLE // Recognise one

$Q = \frac{1}{5}\begin{pmatrix} 3 & -4 \\ 4 & 3 \end{pmatrix}$: columns $\frac{1}{5}(3, 4)$ and $\frac{1}{5}(-4, 3)$ have length 1 and dot product $\frac{1}{25}(-12+12) = 0$. Orthogonal (a rotation, $\det Q = \frac{9+16}{25} = 1$). Check length: $Q(1, 0)^\top = (0.6, 0.8)$, of norm 1 ✓.

6 The spectral theorem

TL;DR // Chapter 6 in one breath

In Lecture 4, diagonalising could fail. For **symmetric** matrices it never fails, and it's even better: the eigenvectors can be chosen **perpendicular and of length 1**. So $A = VDV^\top$ with V orthogonal: no inverse to compute, $V^{-1} = V^\top$.

6.1 Symmetric endomorphisms

DEFINITION // Symmetric endomorphism

In a Euclidean space E, $f \in \mathcal{L}(E)$ is **symmetric** (self-adjoint) if $\langle f(x), y\rangle = \langle x, f(y)\rangle$ for all x, y. On $\mathbb{R}^n$ with the dot product: $x \mapsto Ax$ is symmetric $\iff$ $A^\top = A$ (because $\langle Ax, y\rangle = \langle x, A^\top y\rangle$). Example: orthogonal projections are symmetric (previous section).

THEOREM / PROPOSITION // Spectral theorem

Maps: if f is a symmetric endomorphism of a finite-dimensional Euclidean space E, there is an **orthonormal basis** of E made of eigenvectors of f.
Matrices: if $A \in \mathcal{M}_n(\mathbb{R})$ is symmetric, there exist an **orthogonal** V and a diagonal D with

$$A = VDV^\top,$$

where the diagonal of D holds the n eigenvalues of A, which are all **real**. This is the **spectral decomposition** (A is *orthogonally diagonalisable*).

THEOREM / PROPOSITION // Two facts behind it

For a symmetric A:

1. Its eigenvalues are real.
2. Eigenvectors for **different** eigenvalues are **automatically orthogonal**.

RECEIPTS (proof) // fact 2, with the transpose trick

Let $Av = \lambda v$, $Aw = \mu w$, $\lambda \neq \mu$. Then $\lambda\langle v, w\rangle = \langle Av, w\rangle = \langle v, A^\top w\rangle = \langle v, Aw\rangle = \mu\langle v, w\rangle$. So $(\lambda - \mu)\langle v, w\rangle = 0$, and $\lambda \neq \mu$ gives $\langle v, w\rangle = 0$. ■ *(and that's on periodt)*

RECEIPTS (proof) // spectral theorem, the three-step sketch

1. **One real eigenvalue exists.** Over $\mathbb{C}$, χ_A has a root λ, with an eigenvector $v \in \mathbb{C}^n$, $v \neq 0$. Let $\bar{v}$ be its complex conjugate and $s = \bar{v}^\top Av = \lambda\,\bar{v}^\top v = \lambda \sum_i |v_i|^2$. Since A is real and symmetric, $\bar{s} = v^\top A\bar{v} = (v^\top A\bar{v})^\top = \bar{v}^\top A^\top v = \bar{v}^\top Av = s$, so s is real. As $\sum_i |v_i|^2 > 0$ is real, λ is real; then $\det(A - \lambda I) = 0$ gives a *real* eigenvector.
2. **Stable subspaces.** If F is stable by f ($f(F) \subseteq F$), then $F^\perp$ is stable too: for $x \in F^\perp$ and $v \in F$, $\langle f(x), v\rangle = \langle x, f(v)\rangle = 0$ since $f(v) \in F$.
3. **Induction on n.** Take a unit eigenvector e_1; $F = \mathrm{span}(e_1)$ is stable, so $F^\perp$ (dimension $n-1$) is stable, and f restricted to it is still symmetric. By induction it has an orthonormal eigenbasis $(e_2, \ldots, e_n)$. Add e_1. ■ *(and that's on periodt)*

EXAM CHEAT CODE // "Orthogonally diagonalise the symmetric matrix A"

1. Find the eigenvalues (roots of χ_A). They're real.
2. For each eigenvalue, find a basis of the eigenspace (solve $(A-\lambda I)v=0$).
3. **Normalise** each eigenvector (divide by its length).
4. If an eigenspace has dimension ≥ 2, run **Gram–Schmidt inside it**. (Different eigenspaces are already orthogonal: nothing to do between them.)
5. V = these orthonormal eigenvectors as columns, D = matching eigenvalues. Then $A = VDV^\top$. Check $V^\top V = I$.

EXAMPLE // Worked example 1: 2×2

$A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$ (symmetric).

1. $\chi_A = \lambda^2 - 4\lambda + 3 = (\lambda-1)(\lambda-3)$.
2. $E_3 = \operatorname{span}\big((1,1)\big)$, $E_1 = \operatorname{span}\big((1,-1)\big)$. Check orthogonal: $1-1=0$ ✓ (automatic).
3. Normalise: $v_1 = \frac{1}{\sqrt{2}}(1,1)$, $v_2 = \frac{1}{\sqrt{2}}(1,-1)$.
4. $V = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$, $D = \operatorname{diag}(3,1)$, $A = VDV^\top$. Compared with Lecture 4, we never compute V^{-1}: it's $V^\top$.

EXAMPLE // Worked example 2: 3×3 with a repeated eigenvalue

$A = \begin{pmatrix} 4 & 1 & 1 \\ 1 & 4 & 1 \\ 1 & 1 & 4 \end{pmatrix}$: $\chi_A = (\lambda-6)(\lambda-3)^2$ (Lecture 3).

1. $\lambda = 6$: $E_6 = \operatorname{span}\big((1,1,1)\big)$, normalised $v_1 = \frac{1}{\sqrt{3}}(1,1,1)$.
2. $\lambda = 3$: $A-3I$ is the all-ones matrix, so E_3 is the plane $x+y+z=0$, with basis $(1,-1,0)$, $(1,0,-1)$. These two are **not** orthogonal ($\langle\,,\,\rangle = 1$): Gram–Schmidt inside E_3.
3. $v_2 = \frac{1}{\sqrt{2}}(1,-1,0)$; $u_3 = (1,0,-1) - \frac{1}{2}(1,-1,0) = (\frac{1}{2},\frac{1}{2},-1)$, so $v_3 = \frac{1}{\sqrt{6}}(1,1,-2)$.
4. $V = (v_1\ v_2\ v_3)$, $D = \operatorname{diag}(6,3,3)$, $A = VDV^\top$. Check: $v_1 \perp v_2$, $v_1 \perp v_3$ (sums are 0), $v_2 \perp v_3$: $\frac{1-1+0}{\sqrt{12}} = 0$ ✓.

ML SIDE QUEST // PCA = spectral theorem on the covariance matrix

A covariance matrix Σ is symmetric, so $\Sigma = U\Lambda U^\top$ with U orthogonal. The columns of U are the **principal directions** (perpendicular to each other), and Λ holds the variance along each one. That's exactly what PCA computes.

6.2 Rayleigh quotients

DEFINITION // Quadratic form, Rayleigh quotient

For a symmetric $A \in \mathbb{R}^{n\times n}$: $x^\top Ax$ is a **quadratic form**, and for $x \neq 0$ the **Rayleigh quotient** is

$$R_A(x) = \frac{x^\top Ax}{x^\top x}.$$

THEOREM / PROPOSITION // Properties of the Rayleigh quotient

(i) **Scale invariance:** $R_A(\alpha x) = R_A(x)$ for $\alpha \neq 0$ (only the direction matters).

(ii) If x is an eigenvector for λ, then $R_A(x) = \lambda$.

(iii) **Bounds:** $\lambda_{\min}(A) \leq R_A(x) \leq \lambda_{\max}(A)$ for all $x \neq 0$, with equality iff x is in the eigenspace of $\lambda_{\min}$ (resp. $\lambda_{\max}$). Equivalently, for unit x: $\lambda_{\min} \leq x^\top Ax \leq \lambda_{\max}$.

RECEIPTS (proof) //

(i) $\frac{\alpha^2 x^\top Ax}{\alpha^2 x^\top x} = R_A(x)$. (ii) $\frac{x^\top(\lambda x)}{x^\top x} = \lambda$. (iii) Spectral theorem: orthonormal eigenbasis $v_1, \ldots, v_n$. Write $x = \sum_i \alpha_i v_i$. Then $x^\top Ax = \sum_i \lambda_i \alpha_i^2$ and $x^\top x = \sum_i \alpha_i^2$. So $R_A(x)$ is a weighted average of the λ_i (weights $\alpha_i^2 \geq 0$): it lies between the smallest and largest. ■ *(and that's on periodt)*

EXAMPLE // Rayleigh quotient in numbers

$A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$, eigenvalues 1 and 3.

1. $x = (1, 0)$: $x^\top Ax = 2$, $x^\top x = 1$, $R = 2$.
2. $x = (1, 2)$: $Ax = (4, 5)$, $x^\top Ax = 4 + 10 = 14$, $x^\top x = 5$, $R = 2.8$.
3. $x = (1, 1)$: $R = 3$ (eigenvector of $\lambda_{\max}$). All values lie in $[1, 3]$ ✓.

So $\lambda_{\max} = \max_{\|x\|=1} x^\top Ax$: the top eigenvector is the direction that maximises the quadratic form. That's literally the PCA objective (maximise variance $u^\top \Sigma u$ over unit u).

7 Positive (semi-)definite matrices

DEFINITION // Positive definite / semi-definite

A matrix $A \in \mathbb{R}^{n\times n}$ is

- **positive semi-definite** (PSD, $A \succeq 0$) if A is **symmetric** and $x^\top Ax \geq 0$ for all x;
- **positive definite** (PD, $A \succ 0$) if A is **symmetric** and $x^\top Ax > 0$ for all $x \neq 0$.

Similarly, negative definite: $x^\top Ax < 0$ for all $x \neq 0$. If $x^\top Ax$ takes both signs, A is *indefinite*.

THEOREM / PROPOSITION // The eigenvalue test

A symmetric matrix is PSD $\iff$ all its eigenvalues are ≥ 0, and PD $\iff$ all its eigenvalues are > 0. So PD matrices are invertible (0 is not an eigenvalue); PSD ones may not be.

RECEIPTS (proof) //

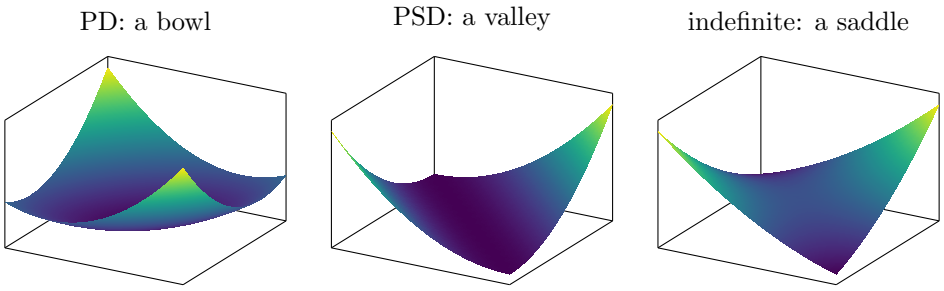
With an orthonormal eigenbasis and $x = \sum_i \alpha_i v_i$: $x^\top Ax = \sum_i \lambda_i \alpha_i^2$. If all $\lambda_i \geq 0$ this is ≥ 0. Conversely, taking $x = v_j$ gives $x^\top Ax = \lambda_j$, so $\lambda_j \geq 0$ is needed. Same with strict inequalities for PD. ■ *(and that's on periodt)*

EXAM CHEAT CODE // "Is A positive (semi-)definite?"

1. **Symmetric?** If not, the answer is no (by definition).
2. **Eigenvalues:** all $> 0 \Rightarrow$ PD; all $\geq 0 \Rightarrow$ PSD; mixed signs $\Rightarrow$ indefinite.
3. 2×2 **shortcut:** $\begin{pmatrix} a & b \\ b & c \end{pmatrix}$ is PD iff $a > 0$ and $ac - b^2 > 0$ (both eigenvalues positive iff their sum tr and product det are positive).
4. **Or write $x^\top Ax$ as a sum of squares** (completing the square).
5. **To show NOT PSD:** exhibit one x with $x^\top Ax < 0$. A negative diagonal entry $a_{ii} < 0$ is an instant counterexample ($x = e_i$).

EXAMPLE // Three verdicts

1. $\begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}$: $x^\top Ax = 2x^2 - 2xy + 2y^2 = x^2 + y^2 + (x-y)^2 > 0$ for $(x, y) \neq 0$. **PD** (eigenvalues $1, 3$).
2. $\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$: $x^\top Ax = (x+y)^2 \geq 0$, but $= 0$ at $(1, -1)$. **PSD, not PD** (eigenvalues $0, 2$).
3. $\begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}$: at $x = (1, -1)$, $x^\top Ax = 1 - 4 + 1 = -2 < 0$. **Indefinite** (eigenvalues $3, -1$).



The graph of $x^\top Ax$ for the three matrices above. PD = strict bowl with a unique bottom (great for optimisation); PSD = flat valley along the kernel; indefinite = saddle.

THEOREM / PROPOSITION // Gram matrices are PSD

For any $A \in \mathbb{R}^{m \times n}$, the **Gram matrix** $G = A^\top A$ is symmetric and PSD. Moreover $x^\top Gx = 0 \iff Ax = 0$.

RECEIPTS (proof) //

$G^\top = A^\top (A^\top)^\top = A^\top A = G$. And $x^\top Gx = x^\top A^\top Ax = (Ax)^\top (Ax) = \|Ax\|_2^2 \geq 0$, zero iff $Ax = 0$, i.e. $x \in \ker A$. ■ *(and that's on periodt)*

THEOREM / PROPOSITION // Making PSD into PD

If A is PSD and $\varepsilon > 0$, then $A+\varepsilon I_n$ is PD. In particular $A^\top A+\varepsilon I$ is always PD, hence invertible.

RECEIPTS (proof) //

For $x \neq 0$: $x^\top(A+\varepsilon I)x = \underbrace{x^\top Ax}_{\geq 0} + \underbrace{\varepsilon\|x\|_2^2}_{>0} > 0$. (Eigenvalues shift from $\lambda_i \geq 0$ to $\lambda_i + \varepsilon > 0$.)

■ *(and that's on periodt)*

ML SIDE QUEST // Ridge regression is this trick

Least squares needs $(X^\top X)^{-1}$, which fails when features are redundant ($X^\top X$ only PSD). **Ridge regression** uses $\hat{\beta} = (X^\top X + \lambda I)^{-1}X^\top y$: always invertible thanks to the theorem above, and it's exactly the $\|w\|_2^2$ regulariser of Section 4. Covariance matrices $\frac{1}{m}X^\top X$ are Gram matrices, hence PSD: variances are never negative.

THEOREM / PROPOSITION // Polar decomposition (bonus)

Every square A can be written $A = RF$ with R orthogonal and F symmetric PSD (one can take $F = (A^\top A)^{1/2}$). If A is invertible, R is unique. *Matrix version of $z = e^{i\theta}|z|$: a rotation times a stretch.*

8 Singular value decomposition (SVD)

TL;DR // Chapter 8 in one breath

The spectral theorem only works for symmetric square matrices. The SVD works for **every** matrix, even rectangular: $A = U\Sigma V^\top$ = rotate, stretch along the axes, rotate. The stretch factors $\sigma_i \geq 0$ are the **singular values**.

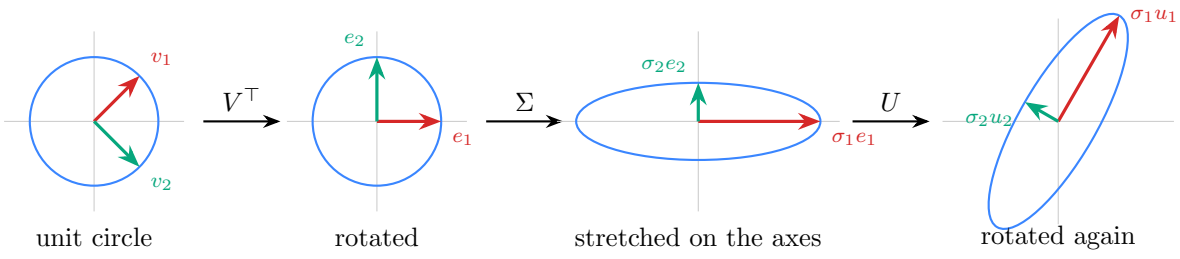
THEOREM / PROPOSITION // Singular value decomposition

Let $A \in \mathbb{R}^{m\times n}$. There exist orthogonal $U \in \mathbb{R}^{m\times m}$, $V \in \mathbb{R}^{n\times n}$, and $\Sigma \in \mathbb{R}^{m\times n}$ with $\Sigma_{ii} = \sigma_i \geq 0$ and zeros elsewhere, such that

$$A = U\Sigma V^\top, \qquad \sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_{\min(m,n)} \geq 0.$$

The σ_i are the **singular values**; the columns u_i of U are the **left singular vectors**, the columns v_i of V the **right singular vectors**. Exactly $r = \operatorname{rank} A$ singular values are non-zero, and

$$A = \sum_{i=1}^{r} \sigma_i\, u_i v_i^\top \qquad \text{(a sum of } r \text{ rank-one pieces).}$$



Read $A = U\Sigma V^\top$ right to left: $V^\top$ rotates (or reflects) the input space, Σ stretches along the axes (the unit circle becomes an ellipse with semi-axes σ_i), U rotates in the output space. $Av_i = \sigma_i u_i$: right singular vectors are the main *input* directions, left singular vectors the main *output* directions, σ_i the gains.

THEOREM / PROPOSITION // SVD and eigen-decompositions

$A^\top A = V(\Sigma^\top\Sigma)V^\top$ and $AA^\top = U(\Sigma\Sigma^\top)U^\top$. So the v_i are eigenvectors of $A^\top A$, the u_i eigenvectors of $AA^\top$, and $\sigma_i = \sqrt{\lambda_i(A^\top A)}$.

RECEIPTS (proof) //

$A^\top A = (U\Sigma V^\top)^\top U\Sigma V^\top = V\Sigma^\top U^\top U\Sigma V^\top = V\Sigma^\top\Sigma V^\top$ since $U^\top U = I$. $\Sigma^\top\Sigma$ is diagonal with entries σ_i^2. Same for $AA^\top$. ■ *(and that's on periodt)*

EXAM CHEAT CODE // "Compute the SVD of A" (small cases)

1. Compute $A^\top A$ (symmetric, PSD).
2. Its eigenvalues $\lambda_1 \geq \lambda_2 \geq \cdots \geq 0$; set $\sigma_i = \sqrt{\lambda_i}$.
3. Orthonormal eigenvectors v_i of $A^\top A$ (spectral theorem): $V = (v_1 \ \cdots \ v_n)$.
4. For each $\sigma_i > 0$: $u_i = \dfrac{Av_i}{\sigma_i}$ (automatically orthonormal). If $m > r$, complete (u_i) into an orthonormal basis of $\mathbb{R}^m$.
5. $\Sigma = m \times n$ with the σ_i on the diagonal. Check $A = U\Sigma V^\top$ (or just $Av_i = \sigma_i u_i$).

EXAMPLE // Worked SVD

$A = \begin{pmatrix} 3 & 0 \\ 4 & 5 \end{pmatrix}$.

1. $A^\top A = \begin{pmatrix} 3 & 4 \\ 0 & 5 \end{pmatrix}\begin{pmatrix} 3 & 0 \\ 4 & 5 \end{pmatrix} = \begin{pmatrix} 25 & 20 \\ 20 & 25 \end{pmatrix}$.
2. $\chi = \lambda^2 - 50\lambda + 225 = (\lambda - 45)(\lambda - 5)$: $\sigma_1 = \sqrt{45} = 3\sqrt{5} \approx 6.71$, $\sigma_2 = \sqrt{5} \approx 2.24$.
3. Eigenvectors: $v_1 = \frac{1}{\sqrt{2}}(1,1)$ for 45, $v_2 = \frac{1}{\sqrt{2}}(1,-1)$ for 5.
4. $u_1 = \dfrac{Av_1}{\sigma_1} = \dfrac{\frac{1}{\sqrt{2}}(3,9)}{3\sqrt{5}} = \frac{1}{\sqrt{10}}(1,3)$, $u_2 = \dfrac{Av_2}{\sigma_2} = \dfrac{\frac{1}{\sqrt{2}}(3,-1)}{\sqrt{5}} = \frac{1}{\sqrt{10}}(3,-1)$. Check $u_1 \perp u_2$: $3 - 3 = 0$ ✓.
5. $A = \underbrace{\frac{1}{\sqrt{10}}\begin{pmatrix} 1 & 3 \\ 3 & -1 \end{pmatrix}}_{U}\underbrace{\begin{pmatrix} 3\sqrt{5} & 0 \\ 0 & \sqrt{5} \end{pmatrix}}_{\Sigma}\underbrace{\frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}}_{V^\top}$ (verified by computer algebra).
6. Best rank-1 approximation: $A_1 = \sigma_1 u_1 v_1^\top = \frac{3}{2}\begin{pmatrix} 1 & 1 \\ 3 & 3 \end{pmatrix} = \begin{pmatrix} 1.5 & 1.5 \\ 4.5 & 4.5 \end{pmatrix}$, error $\|A - A_1\|_F = \sigma_2 = \sqrt{5}$.
7. Bonus: $\|A\|_2 = \sigma_1 = 3\sqrt{5}$ and $|\det A| = \sigma_1\sigma_2 = 15$ ✓.

RECEIPTS (proof) // why the SVD exists (sketch, via the spectral theorem)

1. $A^\top A$ is symmetric PSD, so it has an orthonormal eigenbasis $v_1, \ldots, v_n$ with eigenvalues $\lambda_1 \geq \cdots \geq \lambda_n \geq 0$. Set $\sigma_i = \sqrt{\lambda_i}$, $V = (v_1 \ \cdots \ v_n)$.

2. For $\sigma_i > 0$, $u_i = Av_i/\sigma_i$. Then $\langle u_i, u_j\rangle = \frac{v_i^\top A^\top A v_j}{\sigma_i\sigma_j} = \frac{\lambda_j\langle v_i, v_j\rangle}{\sigma_i\sigma_j} = \delta_{ij}$: orthonormal. Complete to an orthonormal basis of $\mathbb{R}^m$.
3. If $\sigma_i = 0$: $\|Av_i\|^2 = v_i^\top A^\top A v_i = 0$, so $Av_i = 0$.
4. So $Av_i = \sigma_i u_i$ for all i, i.e. $AV = U\Sigma$, i.e. $A = U\Sigma V^\top$. ■ *(and that's on periodt)*

ML SIDE QUEST // Applications of the SVD

- **Low-rank approximation:** keeping the top k terms $\sum_{i\le k}\sigma_i u_i v_i^\top$ gives the *best* rank-k approximation in Frobenius and spectral norm (image compression, LoRA-style ideas, recommender systems).
- **Pseudo-inverse:** $A^+ = V\Sigma^+ U^\top$ (invert the non-zero σ_i): least squares for any matrix.
- **PCA:** the SVD of the centred data matrix gives the principal directions (v_i) and variances (σ_i^2/m).

ML SIDE QUEST // Spectral norm, Lipschitz constants, robust networks

f is L-**Lipschitz** if $\|f(x)-f(y)\|_2 \le L\|x-y\|_2$: small input changes give small output changes. For a linear layer, $\|Wx - Wy\|_2 = \|W(x-y)\|_2 \le \|W\|_2\|x-y\|_2$, so $L = \|W\|_2 = \sigma_{\max}(W)$. For a network $f(x) = W_L\sigma(\cdots\sigma(W_1x))$ with 1-Lipschitz activations (e.g. ReLU):

$$\mathrm{Lip}(f) \le \prod_{\ell=1}^{L}\|W_\ell\|_2.$$

Keeping every $\sigma_{\max}(W_\ell)$ small makes the network less sensitive to tiny input perturbations (adversarial robustness): that's **spectral normalisation** (stable GAN training) and Parseval networks. Same story as the exploding gradients of Lecture 4, but with singular values.

9 The exercises, fully solved

TL;DR // How to use this section

Try first, then check. I worked out every solution below myself, so every solution here is written from scratch and the computations were checked by computer algebra. Each one starts with the **move** it uses.

9.1 Part 1: Short computations

SOLUTION // Question 1: $\|u+v\|_2^2 = \|u\|_2^2 + 2\langle u, v\rangle + \|v\|_2^2$

Move: write the squared norm as an inner product, then expand.

1. $\|u+v\|_2^2 = \langle u+v, u+v\rangle$ (the Euclidean norm is induced by the dot product).
2. Linearity in the first slot: $= \langle u, u+v\rangle + \langle v, u+v\rangle$.
3. Linearity in the second slot: $= \langle u, u\rangle + \langle u, v\rangle + \langle v, u\rangle + \langle v, v\rangle$.
4. Symmetry ($\langle v, u\rangle = \langle u, v\rangle$): $= \|u\|_2^2 + 2\langle u, v\rangle + \|v\|_2^2$.

Check with $u = (1,0)$, $v = (1,1)$: left $\|(2,1)\|^2 = 5$; right $1 + 2\cdot 1 + 2 = 5$ ✓.

SOLUTION // Question 2: $\langle u, Av\rangle = \langle A^\top u, v\rangle$

For $u \in \mathbb{R}^m$, $v \in \mathbb{R}^n$, $A \in \mathbb{R}^{m\times n}$ (so $Av \in \mathbb{R}^m$ and $A^\top u \in \mathbb{R}^n$: both sides make sense):

$$\langle u, Av\rangle = u^\top(Av) = (u^\top A)v = (A^\top u)^\top v = \langle A^\top u, v\rangle,$$

using associativity of the matrix product and $(A^\top u)^\top = u^\top A$.

SOLUTION // Question 3a: $V = \mathrm{span}\{(1,1)\} \subset \mathbb{R}^2$, find $V^\perp$

$(x,y) \in V^\perp \iff \langle (x,y),(1,1)\rangle = 0 \iff x + y = 0 \iff (x,y) = t(1,-1)$.

$$\boxed{V^\perp = \mathrm{span}\{(1,-1)\}}$$

The perpendicular line. Check: $\dim V + \dim V^\perp = 1 + 1 = 2$ ✓.

SOLUTION // Question 3b: $F = \mathrm{span}\{(1,1,0),(1,0,1)\} \subset \mathbb{R}^3$, compute $F^\perp$

$$(x,y,z) \in F^\perp \iff \begin{cases} x + y = 0 \\ x + z = 0 \end{cases} \iff y = -x,\ z = -x.$$

$$\boxed{F^\perp = \mathrm{span}\{(1,-1,-1)\}}$$

Check: $(1,-1,-1)\cdot(1,1,0) = 0$ and $(1,-1,-1)\cdot(1,0,1) = 0$ ✓; $\dim F + \dim F^\perp = 2 + 1 = 3$ ✓. (In $\mathbb{R}^3$, the cross product $(1,1,0)\times(1,0,1) = (1,-1,-1)$ gives it directly.)

SOLUTION // Question 3c: $F = \mathrm{span}\{\mathbf{1}_n\} \subset \mathbb{R}^n$, compute $F^\perp$ and a basis

$x \in F^\perp \iff \langle x, \mathbf{1}_n\rangle = 0 \iff x_1 + x_2 + \cdots + x_n = 0$. So $F^\perp$ is the hyperplane of vectors whose coordinates sum to 0 ("centred" vectors), of dimension $n-1$.
Basis: x_1 is determined by the others: $x_1 = -(x_2 + \cdots + x_n)$, so

$$x = x_2(-1,1,0,\ldots,0) + x_3(-1,0,1,\ldots,0) + \cdots + x_n(-1,0,\ldots,0,1).$$

Basis: $(e_k - e_1)_{2\le k\le n}$, i.e. $(-1,1,0,\ldots,0)$, $(-1,0,1,0,\ldots,0)$, $\ldots$, $(-1,0,\ldots,0,1)$. These $n-1$ vectors are independent (look at coordinates 2 to n).
ML note: centring data, $x \mapsto x - \bar{x}\mathbf{1}_n$, is exactly the orthogonal projection onto this $F^\perp$.

9.2 Part 2: Short proofs

SOLUTION // Question 4: properties of the dot product on $\mathbb{R}^n$

Let $x, y, z \in \mathbb{R}^n$, $\lambda \in \mathbb{R}$, $\langle x, y\rangle = \sum_{i=1}^n x_i y_i$. Each property reduces to a property of real numbers, coordinate by coordinate.

1. **Symmetry:** $\langle x, y\rangle = \sum_i x_i y_i = \sum_i y_i x_i = \langle y, x\rangle$ (multiplication of reals is commutative).
2. **Homogeneity:** $\langle \lambda x, y\rangle = \sum_i (\lambda x_i) y_i = \lambda \sum_i x_i y_i = \lambda\langle x, y\rangle$.
3. **Linearity, first argument:** $\langle x+y, z\rangle = \sum_i (x_i+y_i) z_i = \sum_i x_i z_i + \sum_i y_i z_i = \langle x, z\rangle + \langle y, z\rangle$.
4. **Linearity, second argument:** $\langle x, y+z\rangle = \langle y+z, x\rangle = \langle y, x\rangle + \langle z, x\rangle = \langle x, y\rangle + \langle x, z\rangle$, using 1 and 3.

SOLUTION // Question 5: properties of $F^\perp$ (E finite-dimensional)

(i) $F^\perp$ is a subspace. $0 \in F^\perp$ since $\langle 0, v\rangle = 0$. If $x, y \in F^\perp$, $\lambda \in \mathbb{R}$, then for every $v \in F$: $\langle \lambda x + y, v\rangle = \lambda\langle x, v\rangle + \langle y, v\rangle = 0 + 0 = 0$. So $\lambda x + y \in F^\perp$.
(ii) $F \cap F^\perp = \{0_E\}$. If $x \in F$ and $x \in F^\perp$, then x is orthogonal to every vector of F, including itself: $\langle x, x\rangle = 0$. Positive-definiteness gives $x = 0_E$.
(iii) $\dim F + \dim F^\perp = \dim E$. Let $k = \dim F$, $n = \dim E$. Take an orthonormal basis $(e_1, \dots, e_k)$ of F and complete it into an orthonormal basis $(e_1, \dots, e_n)$ of E (complete into any basis, then Gram–Schmidt; the first k vectors don't change). Write $x = \sum_{i=1}^n \langle x, e_i\rangle e_i$. Then

$$x \in F^\perp \iff \langle x, e_1\rangle = \cdots = \langle x, e_k\rangle = 0 \iff x \in \operatorname{span}(e_{k+1}, \dots, e_n).$$

So $F^\perp = \operatorname{span}(e_{k+1}, \dots, e_n)$ has dimension $n - k$. (With (ii): $E = F \oplus F^\perp$.)
(iv) $(F^\perp)^\perp = F$. First $F \subseteq (F^\perp)^\perp$: if $v \in F$, then $\langle v, x\rangle = 0$ for every $x \in F^\perp$ (by definition of $F^\perp$), so $v \in (F^\perp)^\perp$. Then compare dimensions using (iii) twice: $\dim(F^\perp)^\perp = n - \dim F^\perp = n - (n - k) = k = \dim F$. A subspace contained in another with the same finite dimension is equal to it.

SOLUTION // Question 6: Bessel, $\|p_F(x)\| \leq \|x\|$ with equality iff $x \in F$

Move: Pythagoras.

1. Split $x = p_F(x) + \big(x - p_F(x)\big)$ with $p_F(x) \in F$ and $x - p_F(x) \in F^\perp$ (definition of the orthogonal projection).
2. These two pieces are orthogonal, so by Pythagoras: $\|x\|^2 = \|p_F(x)\|^2 + \|x - p_F(x)\|^2$.
3. The last term is ≥ 0, hence $\|p_F(x)\|^2 \leq \|x\|^2$, and $\|p_F(x)\| \leq \|x\|$ (both are ≥ 0).
4. Equality $\iff$ $\|x - p_F(x)\| = 0$ $\iff$ $x = p_F(x)$. If $x = p_F(x)$ then $x \in F$; conversely, if $x \in F$, then x itself satisfies the two defining conditions ($x \in F$, $x - x = 0 \in F^\perp$), so $p_F(x) = x$.

SOLUTION // Question 7: characterising orthogonal projectors

Let $p \circ p = p$, $F = \operatorname{range}(p)$. Useful remark: for $z \in F$, $z = p(w)$ for some w, so $p(z) = p(p(w)) = p(w) = z$: p **fixes every vector of** F.
(i) $\Rightarrow$ (ii). If p is the orthogonal projection onto F: $p(x) \in F$ and $y - p(y) \in F^\perp$, so $\langle p(x), y - p(y)\rangle = 0$.
(ii) $\Rightarrow$ (iii). From (ii): $\langle p(x), y\rangle = \langle p(x), p(y)\rangle$. Applying (ii) with x and y swapped: $\langle p(y), x - p(x)\rangle = 0$, i.e. $\langle x, p(y)\rangle = \langle p(x), p(y)\rangle$. Both equal $\langle p(x), p(y)\rangle$, so $\langle p(x), y\rangle = \langle x, p(y)\rangle$.
(iii) $\Rightarrow$ (i). Fix y. We know $p(y) \in F$; we show $y - p(y) \in F^\perp$. Take any $z \in F$; then $z = p(z)$, so

$$\langle z, y - p(y)\rangle = \langle p(z), y\rangle - \langle p(z), p(y)\rangle \overset{\text{(iii)}}{=} \langle p(z), y\rangle - \langle p(p(z)), y\rangle = \langle p(z), y\rangle - \langle p(z), y\rangle = 0.$$

So $p(y)$ satisfies both conditions: $p = p_F$.
And $\ker p = F^\perp$ (the hint). If $p(x) = 0$ then $x = x - p(x) \in F^\perp$. Conversely if $x \in F^\perp$: $p(x) \in F$ and $p(x) = x - (x - p(x))$ is a difference of two vectors of $F^\perp$, so $p(x) \in F \cap F^\perp = \{0\}$. With $E = F \oplus F^\perp$: p keeps the F-part of every vector and kills the $F^\perp$-part.

9.3 Part 3: Exercises

SOLUTION // Exercise 1: a weighted inner product on functions

Show that $\varphi(f,g) = \int_{-1}^{1} \sqrt{1-x^2}\, f(x)g(x)\, dx$ *is an inner product on* $E = C([-1,1],\mathbb{R})$.
Well defined: $x \mapsto \sqrt{1-x^2}f(x)g(x)$ is continuous on $[-1,1]$, so the integral exists.

1. **Symmetry:** $f(x)g(x) = g(x)f(x)$, so $\varphi(f,g) = \varphi(g,f)$.
2. **Linearity in f:** $\varphi(\alpha f_1 + \beta f_2, g) = \int \sqrt{1-x^2}(\alpha f_1 + \beta f_2)g = \alpha\varphi(f_1,g) + \beta\varphi(f_2,g)$ by linearity of the integral. By symmetry, also linear in g.
3. **Positivity:** $\varphi(f,f) = \int_{-1}^{1} \underbrace{\sqrt{1-x^2}}_{\geq 0} \underbrace{f(x)^2}_{\geq 0} dx \geq 0$.
4. **Definiteness:** assume $\varphi(f,f) = 0$. The function $h(x) = \sqrt{1-x^2}f(x)^2$ is continuous, ≥ 0, with integral 0 on $[-1,1]$, hence $h = 0$ on $[-1,1]$. For $x \in]-1,1[$, $\sqrt{1-x^2} > 0$, so $f(x)^2 = 0$, i.e. $f(x) = 0$. At $x = \pm 1$ the weight vanishes, but f is continuous: $f(1) = \lim_{x\to 1^-} f(x) = 0$, same at -1. So $f = 0$.

Hence φ is an inner product. *The only subtle step is the endpoints, where the weight is 0: continuity saves us.*

SOLUTION // Exercise 2: $\left\|\sum_{i=1}^n \alpha_i x_i\right\|^2 \leq \left(\sum_{i=1}^n \alpha_i^2\right)\left(\sum_{i=1}^n \|x_i\|^2\right)$

Moves: triangle inequality in E, then Cauchy–Schwarz in $\mathbb{R}^n$. Careful: E is only a *normed* space (no inner product on E), so Cauchy–Schwarz must be used on numbers.

1. Triangle inequality and homogeneity in E:
$$\left\|\sum_{i=1}^n \alpha_i x_i\right\| \leq \sum_{i=1}^n \|\alpha_i x_i\| = \sum_{i=1}^n |\alpha_i|\, \|x_i\|.$$
2. The right side is the dot product in $\mathbb{R}^n$ of $a = (|\alpha_1|,\dots,|\alpha_n|)$ and $b = (\|x_1\|,\dots,\|x_n\|)$. Cauchy–Schwarz in $\mathbb{R}^n$: $\langle a,b\rangle^2 \leq \|a\|_2^2\|b\|_2^2$, i.e.
$$\left(\sum_{i=1}^n |\alpha_i|\|x_i\|\right)^2 \leq \left(\sum_{i=1}^n \alpha_i^2\right)\left(\sum_{i=1}^n \|x_i\|^2\right).$$
3. Both sides of step 1 are ≥ 0, so squaring keeps the inequality. Chain steps 1 and 2. ■ *(and that's on periodt)*

SOLUTION // Bonus: $S_n(\mathbb{R})$ and $A_n(\mathbb{R})$ are orthogonal complements

$(M \mid N) = \operatorname{tr}(M^\top N)$ on $\mathcal{M}_n(\mathbb{R})$.
1. Inner product. First compute: $(M^\top N)_{jj} = \sum_i m_{ij}n_{ij}$, so $\operatorname{tr}(M^\top N) = \sum_{i,j} m_{ij}n_{ij}$ (the dot product of the two matrices read as vectors of n^2 entries).

- Symmetry: $\sum_{i,j} m_{ij}n_{ij} = \sum_{i,j} n_{ij}m_{ij}$. (Or: $\operatorname{tr}(M^\top N) = \operatorname{tr}\left((M^\top N)^\top\right) = \operatorname{tr}(N^\top M)$.)
- Bilinearity: the trace and the matrix product are linear.
- Positive-definite: $(M \mid M) = \sum_{i,j} m_{ij}^2 \geq 0$, and $= 0$ iff all $m_{ij} = 0$, i.e. $M = 0$.

2. Norm. An inner product always induces a norm (Section 4): $\|M\| = \sqrt{\operatorname{tr}(M^\top M)} = \sqrt{\sum_{i,j} m_{ij}^2}$. It's the Frobenius norm.

3. Subspaces. $S_n = \{M \mid M^\top = M\}$, $A_n = \{M \mid M^\top = -M\}$. Both contain 0; for $M, N \in S_n$: $(\lambda M + N)^\top = \lambda M^\top + N^\top = \lambda M + N$. Same computation for A_n with minus signs. (Or: $S_n = \ker(M \mapsto M^\top - M)$, $A_n = \ker(M \mapsto M^\top + M)$.)

4. Complementary and orthogonal.

- *Sum:* every $M = \underbrace{\frac{1}{2}(M + M^\top)}_{\in S_n} + \underbrace{\frac{1}{2}(M - M^\top)}_{\in A_n}$.
- *Intersection:* if $M^\top = M$ and $M^\top = -M$, then $M = -M$, so $M = 0$. Hence $\mathcal{M}_n = S_n \oplus A_n$.
- *Orthogonal:* take $S \in S_n$, $A \in A_n$. Using $\operatorname{tr}(XY) = \operatorname{tr}(YX)$ and $\operatorname{tr}(X^\top) = \operatorname{tr}(X)$:

  $$(S \mid A) = \operatorname{tr}(S^\top A) = \operatorname{tr}(SA), \qquad (S \mid A) = (A \mid S) = \operatorname{tr}(A^\top S) = -\operatorname{tr}(AS) = -\operatorname{tr}(SA).$$

  So $(S \mid A) = -(S \mid A)$, hence 0. Then $A_n \subseteq S_n^\perp$, and dimensions (next question) give $A_n = S_n^\perp$.

5. Dimensions and bases. With E_{ij} = the matrix with a 1 at (i, j) and 0 elsewhere:

- S_n: basis $\{E_{ii}\}_{1 \le i \le n} \cup \{E_{ij} + E_{ji}\}_{i<j}$; a symmetric matrix is decided by its diagonal and upper triangle. $\dim S_n = n + \frac{n(n-1)}{2} = \frac{n(n+1)}{2}$.
- A_n: the diagonal is 0 ($a_{ii} = -a_{ii}$); basis $\{E_{ij} - E_{ji}\}_{i<j}$. $\dim A_n = \frac{n(n-1)}{2}$.
- Check: $\frac{n(n+1)}{2} + \frac{n(n-1)}{2} = n^2 = \dim \mathcal{M}_n$ ✓. Example $n = 2$: S_2 has basis $\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, A_2 has basis $\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$; $3 + 1 = 4$.

10 Final boss: the cheat sheet

10.1 "If they ask you this, do that"

If asked...	Do this
Show φ is an inner product	Symmetry, linearity in slot 1 (+ symmetry), $\varphi(x,x) \geq 0$, then $\varphi(x,x) = 0 \Rightarrow x = 0$ (continuity for integrals).
Show N is a norm	$N(x) = 0 \iff x = 0$; $N(\alpha x) = \lvert\alpha\rvert N(x)$; triangle inequality. If $N = \sqrt{\text{inner product}}$: automatic.
Expand $\lVert x \pm y\rVert^2$	$\lVert x\rVert^2 \pm 2\langle x, y\rangle + \lVert y\rVert^2$.
Prove an inequality with sums or norms	Triangle inequality, then Cauchy–Schwarz $\lvert\langle x, y\rangle\rvert \leq \lVert x\rVert\lVert y\rVert$ (in $\mathbb{R}^n$ if E has no inner product).
Show vectors are independent	If they're orthogonal and non-zero: dot with v_j, done.
$F^\perp$	One equation $\langle x, f_i\rangle = 0$ per spanning vector; solve; check $\dim F + \dim F^\perp = \dim E$.
Orthonormal basis	Gram–Schmidt: subtract projections on previous e_i, normalise.
Project x onto F	Orthonormal basis e_i of F; $p_F(x) = \sum\langle x, e_i\rangle e_i$; check $x - p_F(x) \perp F$.
Distance from x to F	$\lVert x - p_F(x)\rVert$. Hyperplane with normal n: $\lvert\langle x, n\rangle\rvert / \lVert n\rVert$.
Is Q orthogonal?	Columns unit length and pairwise orthogonal $\iff Q^\top Q = I$.
Orthogonally diagonalise symmetric A	Eigenvalues; eigenvectors; normalise; Gram–Schmidt inside repeated eigenspaces; $A = VDV^\top$.
PSD / PD?	Symmetric? Eigenvalues ≥ 0 / > 0. 2×2: $a > 0$ and det > 0 for PD. Not PSD: one x with $x^\top Ax < 0$.
Gram matrix $A^\top A$	Symmetric PSD, $x^\top A^\top Ax = \lVert Ax\rVert^2$. $A^\top A + \varepsilon I$ is PD.
SVD	Eigen-decompose $A^\top A$: $\sigma_i = \sqrt{\lambda_i}$, v_i eigenvectors, $u_i = Av_i/\sigma_i$.
Spectral norm $\lVert A\rVert_2$	$\sigma_{\max}(A) = \sqrt{\lambda_{\max}(A^\top A)}$. For symmetric A: $\max\lvert\lambda_i\rvert$.

10.2 Formula sheet

Dot product	$\langle x, y\rangle = \sum x_i y_i = x^\top y = \lVert x\rVert_2\lVert y\rVert_2 \cos\theta; \quad \langle Ax, y\rangle = \langle x, A^\top y\rangle$
Norms on $\mathbb{R}^n$	$\lVert x\rVert_1 = \sum\lvert x_i\rvert$, $\lVert x\rVert_2 = \sqrt{\sum x_i^2}$, $\lVert x\rVert_p = (\sum\lvert x_i\rvert^p)^{1/p}$, $\lVert x\rVert_\infty = \max\lvert x_i\rvert$
Key identities	$\lVert x + y\rVert^2 = \lVert x\rVert^2 + 2\langle x, y\rangle + \lVert y\rVert^2$; $4\langle x, y\rangle = \lVert x + y\rVert^2 - \lVert x - y\rVert^2$
Inequalities	$\lvert\langle x, y\rangle\rvert \leq \lVert x\rVert\lVert y\rVert$; $\lVert x + y\rVert \leq \lVert x\rVert + \lVert y\rVert$; $\lVert p_F(x)\rVert \leq \lVert x\rVert$
Orthonormal basis	$x = \sum\langle x, e_i\rangle e_i$, $\lVert x\rVert^2 = \sum\langle x, e_i\rangle^2$
Complement	$F \cap F^\perp = \{0\}$, $\dim F + \dim F^\perp = \dim E$, $(F^\perp)^\perp = F$
Projection	$p_F(x) = \sum_{i\leq k}\langle x, e_i\rangle e_i$; $x - p_F(x) \in F^\perp$; $d(x, F) = \lVert x - p_F(x)\rVert$
Orthogonal matrix	$Q^\top Q = QQ^\top = I$, $Q^{-1} = Q^\top$, $\det Q = \pm 1$, $\lVert Qx\rVert = \lVert x\rVert$
Spectral theorem	$A = A^\top \Rightarrow A = VDV^\top$, V orthogonal, D real
Rayleigh	$\lambda_{\min} \leq \dfrac{x^\top Ax}{x^\top x} \leq \lambda_{\max}$
PSD	$x^\top Ax \geq 0 \iff$ all $\lambda_i \geq 0$ (symmetric A)
SVD	$A = U\Sigma V^\top = \sum_{i\leq r}\sigma_i u_i v_i^\top$, $\sigma_i = \sqrt{\lambda_i(A^\top A)}$, $Av_i = \sigma_i u_i$
Matrix norms	$\lVert A\rVert_F = \sqrt{\sum a_{ij}^2}$; $\lVert A\rVert_1 =$ max col sum; $\lVert A\rVert_\infty =$ max row sum; $\lVert A\rVert_2 = \sigma_{\max} \leq \lVert A\rVert_F$

10.3 Hall of Ls

COMMON L (don't do this) // Do not

1. Forget definiteness when proving an inner product: "$\varphi(x,x) \geq 0$" is not enough, you need "$= 0 \Rightarrow x = 0$".
2. Use the projection formula $\sum \langle x, e_i \rangle e_i$ with a basis that isn't orthonormal.
3. Write $\|x+y\| = \|x\| + \|y\|$. That's only an inequality (equality only when they point the same way).
4. Write $\|x+y\|^2 = \|x\|^2 + \|y\|^2$ without checking $x \perp y$.
5. Use Cauchy–Schwarz in a space that only has a norm. It needs an inner product (use it in $\mathbb{R}^n$ on the numbers instead).
6. Call a non-symmetric matrix PSD because $x^\top A x \geq 0$. The definition requires symmetry.
7. Assume eigenvectors of a repeated eigenvalue are orthogonal. Only eigenvectors of *different* eigenvalues are automatically orthogonal: Gram–Schmidt the rest.
8. Confuse $\|A\|_2$ (largest singular value) with $\|A\|_F$ (all entries squared).
9. Confuse singular values and eigenvalues: $\sigma_i = \sqrt{\lambda_i(A^\top A)}$, always ≥ 0; equal to $|\lambda_i(A)|$ only for symmetric A.
10. Write V^{-1} in a spectral decomposition and compute it the long way. It's $V^\top$.

10.4 The whole lecture in 10 lines

TL;DR // Screenshot this

1. Inner product: symmetric, bilinear, positive-definite. Dot product, weighted dot product, $\int fg$, $\operatorname{tr}(A^\top B)$.
2. Norm: positive, homogeneous, triangle inequality. $\|x\| = \sqrt{\langle x, x \rangle}$; L^1, L^2, L^∞ with diamond / circle / square unit balls.
3. Metric: positive, symmetric, triangle inequality. Choosing it changes nearest neighbours.
4. $x \perp y \iff \langle x, y \rangle = 0$. Pythagoras. Cauchy–Schwarz $|\langle x, y \rangle| \leq \|x\|\|y\| \Rightarrow$ cosine similarity in $[-1, 1]$.
5. Orthonormal basis: coordinates are dot products. Gram–Schmidt builds one.
6. $F^\perp$: everything perpendicular to F; $E = F \oplus F^\perp$; $(F^\perp)^\perp = F$.
7. Orthogonal projection $p_F(x) = \sum \langle x, e_i \rangle e_i =$ closest point of F (least squares, PCA).
8. Spectral theorem: symmetric $\Rightarrow A = VDV^\top$ with orthonormal eigenvectors and real eigenvalues.
9. PSD $\iff$ eigenvalues ≥ 0; Gram matrices $A^\top A$ are PSD; $+\varepsilon I$ makes them PD (ridge).
10. SVD: every matrix $= U\Sigma V^\top$ (rotate, stretch, rotate); best low-rank approximations; $\|A\|_2 = \sigma_{\max}$ controls Lipschitz constants.

10.5 The A4 sheet

EXAM CHEAT CODE // Building a one-page A4 cheat sheet

- **Goal:** a single A4 sheet (both sides) holding everything from Lectures 1 to 5. Keep it readable: definitions, recipes and counterexamples, nothing else.
- **Write it by hand**: copying the formulas out is already half the revision.

What to put on the A4 sheet: the "If they ask you this, do that" tables and formula sheets of Lectures 1 to 5, plus the counterexamples from each Hall of Ls ($\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$, $\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix}$, $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$, $\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}\dots$). They catch most true/false questions.

That's the linear algebra part done. Go get that exam, bestie.